

DATA QUALITY REPORT

Financial Sales Data Analysis Project

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Tools Used: Python (Pandas, NumPy), Jupyter Notebook
Project Type: Data Cleaning, Data Quality Assessment & Financial Dashboard Development

1. Executive Summary

This report presents the data quality assessment and cleaning process performed on the Financial Sales Dataset. The primary objective was to identify and resolve data quality issues that could affect business analysis, KPI calculations, and dashboard accuracy.

The assessment focused on missing values, duplicate records, data type validation, and overall dataset consistency. After applying appropriate cleaning techniques, the dataset was transformed into a reliable and analysis-ready dataset suitable for dashboard development and business reporting.

2. Dataset Overview

Attribute	Details
Dataset Name	Advanced Financial Messy Dataset
Total Records (Before Cleaning)	1,280
Total Columns	13
Project Domain	Financial Sales Analysis
Objective	Prepare a high-quality dataset for business analysis and dashboard reporting

3. Data Quality Assessment

Overall Data Quality Summary

Data Quality Check	Result	Status
Missing Values	381 missing values identified	✓ Resolved
Duplicate Records	68 duplicate records identified	✓ Removed
Data Type Validation	Date converted to DateTime	✓ Validated

Data Quality Check	Result	Status
Data Consistency	Text fields standardized	✔ Completed
Dataset Readiness	Ready for Analysis	✔ Approved

4. Missing Values Analysis

The dataset contained missing values in five columns. Missing data represented approximately 6% of the records in each affected column.

Column Name	Missing Values	Percentage	Treatment
Customer_Age	78	6.09%	Filled using Median
Marketing_Spend	77	6.02%	Filled using Median
Region	77	6.02%	Filled using Mode
Customer_Satisfaction	76	5.94%	Filled using Median
Gender	73	5.70%	Filled using Mode

Missing Value Treatment

- **Median Imputation** was applied to numerical columns because it is resistant to extreme values and preserves the overall distribution.
- **Mode Imputation** was applied to categorical columns to retain the most frequently occurring category and maintain data consistency.

Following the imputation process, **all missing values were successfully resolved**, resulting in a complete dataset for further analysis.

5. Duplicate Records Analysis

Duplicate records can lead to inaccurate business insights by counting the same transaction multiple times.

Metric	Value
Duplicate Records Detected	68
Duplicate Records Removed	68
Remaining Duplicates	0

Action Taken

Duplicate transactions were identified using Python's duplicated() function and permanently removed using the drop_duplicates() method. This ensured that every transaction represented a unique business record.

6. Data Type Validation

Data types were reviewed to ensure that every column was stored in the correct format before performing analysis.

Column Type	Validation
Date	Converted to DateTime format
Revenue	Numeric
Profit	Numeric
Customer Age	Numeric
Customer Satisfaction	Numeric
Region	Text
Gender	Text
Product Category	Text

Correct data types improved calculation accuracy and enabled reliable filtering, grouping, and visualization.

7. Data Cleaning Activities

The following cleaning operations were performed during data preparation:

- Removed **68 duplicate records**.
- Converted the **Date** column into DateTime format.
- Replaced missing values in **Customer_Age** using the median.
- Replaced missing values in **Marketing_Spend** using the median.
- Replaced missing values in **Customer_Satisfaction** using the median.
- Filled missing values in **Region** using the mode.
- Filled missing values in **Gender** using the mode.
- Verified numerical and categorical data types.

- Prepared the dataset for KPI calculations and dashboard visualization.

8. Data Quality Improvements		
Quality Metric	Before Cleaning	After Cleaning
Missing Values	381	0
Duplicate Records	68	0
Invalid Date Format	Present	Corrected
Dataset Ready for Analysis	No	Yes

9. Final Dataset Status

The dataset successfully passed all required data quality checks after cleaning.

- ✔ Missing values eliminated
 - ✔ Duplicate records removed
 - ✔ Data types validated
 - ✔ Dataset standardized
 - ✔ Ready for business analysis
 - ✔ Ready for dashboard visualization
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10. Conclusion

The data quality assessment significantly improved the reliability, consistency, and usability of the Financial Sales Dataset. Missing values were appropriately handled using statistical imputation techniques, duplicate records were removed, and all critical data types were validated.

The final cleaned dataset provides a strong foundation for exploratory data analysis, KPI generation, business insight reporting, and interactive dashboard development. These improvements ensure that business decisions and visualizations are based on accurate and trustworthy data.